

Chapter 145 — Surgical and Radiographic Anatomy of the Adrenals

- The adrenal glands are **paired retroperitoneal organs**, unique in their embryology, anatomy and vital role in homeostasis.
- They produce **mineralocorticoids, glucocorticoids, sex steroids and catecholamines**, which regulate crucial physiological processes throughout the body.
- Knowledge of adrenal anatomy underpins medical and surgical management of adrenal disease and is **essential for the retroperitoneal surgeon**.

ANATOMIC RELATIONSHIPS

- The glands lie **cephalad to the kidneys in the retroperitoneum**; their position differs between right and left.
- Both lie at the **level of the 11th or 12th rib**: the **right gland is higher**, and the left extends inferiorly to the **first lumbar space**.
- Both are **enclosed within the perirenal (Gerota) fascia** and entirely surrounded by perirenal fat; a **thin layer of connective tissue** separates each gland from the upper pole of its kidney.
- Grossly **yellow-orange**, noticeably more orange than the surrounding fat.
- Dimensions **2–3 cm wide and 4–6 cm long**; weight **approximately 5 g each (range 2–6 g)**, with **no difference between the sexes**.

SURGICAL LANDMARKS

- Dorsolaterally, both glands lie against the **crus of the diaphragm**.
- The relationships to surrounding intra-abdominal and retroperitoneal organs **determine the surgical approach**; cross-sectional anatomy (Fig. 145.3) shows the right gland with the **liver and inferior vena cava (IVC)** and the left with the **splenic vasculature and pancreas**, each lying anterior to its diaphragmatic crus.

RIGHT VERSUS LEFT ADRENAL GLAND

Feature	Right	Left
Shape	Triangular	Crescentic
Level	Higher	Lower; extends to first lumbar space
Relation to kidney	Nearly directly cranial to the upper pole	Lateral surface on the medial aspect of the upper pole
Anterior	Underside of liver (anterolateral); duodenum (anteromedial)	Splenic vessels and body of pancreas
Medial	Lateral margin of the IVC	Aorta
Posterior	Psoas muscle	Psoas muscle
Adrenal vein	Short ; enters the posterior aspect of the IVC	Longer ; enters the cranial aspect of the left renal vein , often after joining the inferior phrenic vein
Computed tomography (CT) position	Just superior to the upper pole; posterolateral to the IVC	Slightly more caudal; anteromedial to the upper pole, posterior to the splenic vessels

Feature	Right	Left
Transabdominal exposure (Fig. 145.2)	Liver retracted superiorly, IVC retracted medially, duodenum and kidney drawn down	Pancreas and spleen retracted superiorly, splenic flexure drawn medially

Figure Legends

- **Fig. 145.1 — Dissected glands.** Right: the **IVC is excised to expose the gland**; the coeliac trunk, its branches and the coeliac autonomic plexus are displayed, with autonomic nerves reaching the gland from the inferior phrenic and coeliac ganglia. Left: **several middle adrenal arteries arise from the aorta**, the inferior adrenal artery may arise from a **perinephric (capsular) branch of the renal artery**, and perinephric vessels run within Gerota fascia.
- **Fig. 145.2 — Netter abdominal exposures** of the right and left glands, with an inset cross-section showing capsule, cortex and medulla.

Concept — *Side determines the neighbours and the danger: the right gland is high, hepatic and retrocaval with a short vein into the IVC; the left is lower, behind the pancreas and splenic vessels with a long vein into the renal vein.*

ADRENAL VASCULATURE

- Both arterial and venous anatomy are **highly variable**.

Arterial Supply

THREE ARTERIAL SOURCES OF THE ADRENAL GLAND (FIG. 145.4)

Artery	Usual origin	Rare origins
Superior	Branches of the inferior phrenic artery	Aorta, coeliac axis, intercostal arteries
Middle	Lateral aspect of the aorta	Inferior phrenic artery, renal artery
Inferior	Superior aspect of the ipsilateral renal artery	Perinephric (capsular) branch of the renal artery (Fig. 145.1)

- Each source branches into a **cascade of 10–50 small arteries** that penetrate the capsule.

Intra-adrenal Distribution

THREE PATTERNS OF BLOOD DISTRIBUTION WITHIN THE GLAND (FIGS. 145.5–145.6)

Pattern	Course	Territory
Capsular arteries	Penetrate no deeper than the capsule	Capsule only
Fenestrated cortical sinusoidal capillaries	Perfuse the cortex, then drain into fenestrated medullary sinusoids	Cortex (arterial), then medulla (venous)
Medullary arterioles	Run within the trabeculae, bypassing the cortical bed	Medullary sinusoids (arterial)

- Microvascular sequence (Fig. 145.6): **capsular artery → subcapsular plexus → sinusoidal vessels → deep plexus → medullary plexus → medullary vein**.
- The medulla therefore has a **dual supply: arterial blood** from medullary arterioles and **venous blood** from cortical sinusoids that have already perfused the cortex.
- Cortical venous blood reaching the medulla carries a **high glucocorticoid concentration**, which plays a role in **epinephrine (adrenaline) synthesis**; the dual supply is thus important for medullary catecholamine production.

- The source groups the system into a **cortical supply** (cortical sinusoids draining to medullary capillaries and the medullary vein) and a **medullary supply** (medullary arterioles and cortical sinusoids).

Venous Drainage

- Each gland is drained by a **single central vein exiting anteromedially**; emissary veins connect it to the pericapsular plexus, forming an intricate intraglandular network.
- **Right versus left** (Fig. 145.7): right vein **short, into the posterior IVC** | left vein **longer, into the cranial aspect of the left renal vein**, sometimes first joining the **inferior phrenic vein**.

Concept — *Many arteries in, one vein out: supply is a diffuse cascade from three sources, drainage converges on a single central vein, and the medulla is perfused in series downstream of the cortex, which couples cortical glucocorticoid output to medullary epinephrine synthesis.*

ADRENAL NERVES

- Innervation governs **catecholamine release from medullary chromaffin cells** (Fig. 145.8).
- Pathway: **preganglionic sympathetic fibres** from the lower thoracic and lumbar cord (Fig. 145.8 depicts the intermediolateral cell column at T5–T8) → sympathetic chain → greater, lesser and least thoracic and first lumbar splanchnic nerves → **through the coeliac ganglion without synapse** → plexus at the adrenal capsule → traverse the cortex → **ramify around medullary cells**.
- **Postganglionic fibres from the coeliac ganglion** supply the adrenal blood vessels (Fig. 145.8).
- Medullary secretory products enter the circulation via **fenestrated capillaries**.
- **Cholinergic innervation of the cortex** has been described, but is less well characterised than medullary innervation.

Concept — *The medulla behaves as a modified sympathetic ganglion: it receives preganglionic fibres directly, and its chromaffin cells take the place of postganglionic neurones.*

EMBRYOLOGY

- Adrenal development was hypothesised until **1911**, when anatomy across stages of human development was first described; the original illustration showed an early **fetal zone of immature cortical cells and groups of sympathogonia**.
- The fetal zone is important during development and **shrinks dramatically after birth**.

Developmental Sequence

- **Weeks 3–4**: columns of **coelomic epithelium condense** → over the next 2 weeks they proliferate and migrate to the cranial end of the mesonephros → **adrenogonadal ridge**, common precursor of adrenal and gonad → **cephalic portion forms the adrenal cortex, caudal portion the gonads** (Fig. 145.9).
- **Weeks 8–10**: differentiation into **definitive and fetal zones** → surrounding mesenchyme forms the **capsule** → **neural crest cells migrate medially to form the medulla** → vessels and nerves form.
- **About week 8**: the kidneys **ascend from the pelvis** to meet the adrenals.
- **Birth**: glands relatively large, **twice adult weight**.
- **First year**: **fetal zone involutes**, replaced by the definitive zone, which forms the **zona glomerulosa (ZG) and zona fasciculata (ZF)** under **adrenocorticotrophic hormone (ACTH)** → chromaffin cells scattered in the fetal zone aggregate into the medulla, **homologous to a sympathetic ganglion without postganglionic processes**.
- **Ages 6–8 years**: the **zona reticularis (ZR)**, minor until early childhood, **expands to adult volume**.

Developmental Anomalies

- **Renal agenesis:** the adrenal is still found in its **orthotopic position**, but may be **discoïd** rather than triangular or crescentic.
- **Adrenal rests** — ectopic cortical or medullary tissue:
 - Found in about **1% of adults**, near the adrenals and close to the **coeliac axis**.
 - Along the **path of gonadal descent**: gonad-associated rests in **7.5–15% of neonates**, usually regressing.
 - In **congenital adrenal hyperplasia (CAH)** they may present as **testicular masses**.
 - Clinical relevance: **compensatory hypertrophy after adrenalectomy**, **inadvertent excision of heterotopic tissue** at unrelated surgery, and **neoplastic transformation**.

Figure Legends

- **Fig. 145.10** — cross-sections of the **infant and adult glands**, contrasting the infant cortical arrangement with the adult concentric zones.

Concept — Two origins, one gland: *the cortex is coelomic mesoderm sharing an adrenogonadal ridge with the gonad, the medulla is neural crest, and the kidney develops separately — hence orthotopic adrenals in renal agenesis and rests along the path of gonadal descent.*

HISTOLOGY

- An **outer cortex (about 90% of mass)** and an **inner medulla**, within a **capsule of hypocellular fibrous tissue**.
- The cortex has **three concentric zones**, each identifiable by cell appearance and architecture (Figs. 145.10–145.11).

ADRENAL ZONES — PROPORTION, STRUCTURE AND PRODUCT (FIGS. 145.5, 145.11)

Zone	Proportion	Cells and architecture	Product
ZG (outer)	~15% of cortex	Small polyhedral cells in ovoid clusters and curved columns, enveloped by sinusoidal capillaries; scant cytoplasm, high nuclear-to-cytoplasmic ratio , sparse lipid	Aldosterone (smooth endoplasmic reticulum and mitochondria)
ZF (middle)	~80% of cortex — largest	Large pale polyhedral cells in long straight cords separated by capillaries; most lipid → "clear cells"	Cortisol
ZR (inner)	5–7% of cortex	Smaller cells in anastomosing cords	Dehydroepiandrosterone sulphate (DHEAS)
Medulla	~10% of weight and volume (text); 10–20% (Fig. 145.11)	Chromaffin cells in ovoid clusters and cords; large, epithelioid, poorly outlined, variable nuclei, close to medullary capillaries	Epinephrine, norepinephrine

- The text gives a single value (cortex about 90%, medulla 10%); Fig. 145.11, from a physiology source, gives **ranges of 80–90% and 10–20%**. Fig. 145.5 groups ZF and ZR together as the source of **glucocorticoids and sex hormones**.
- **Chromaffin cells** are **postganglionic sympathetic neurones that have lost their axons and dendrites**; staining and electron microscopy distinguish epinephrine- from norepinephrine-secreting cells, and preganglionic sympathetic terminals end on both (Fig. 145.5).

Concept — Architecture tracks product: *each cortical layer has a distinct cell form and secretion, lipid content peaks where steroid output is greatest (ZF), and the medulla is neural tissue set within an endocrine cortex.*

RADIOLOGY

- Modalities: **CT, magnetic resonance imaging (MRI), ultrasound (US), angiography, metaiodobenzylguanidine (MIBG) scan and positron emission tomography (PET)**; imaging evaluates abnormal morphology or function (covered in later chapters). This section reviews **normal anatomy** on CT, MRI, US and angiography (Fig. 145.12).

Computed Tomography

- **Most widely used modality**; perinephric fat allows the gland to be seen with outstanding resolution. Excellent morphology, but **no functional data**.
- Axial appearance: **inverted V or Y**; limbs **3–6 mm wide**, usually **thinner than the adjacent diaphragmatic crura**. Position of each gland: see the right-versus-left table.
- Unenhanced attenuation **≤10 Hounsfield units (HU)** reflects intracellular lipid — the threshold that identifies **lipid-rich adrenal tissue** on noncontrast CT.

Magnetic Resonance Imaging

- **Spatial resolution slightly inferior to CT**; a useful adjunct in selected pathology, and **preferred where radiation exposure is a concern**.
- **Coronal or sagittal images** confirm the adrenal origin of a mass when axial images are equivocal.
- **Contrast resolution (T1- and T2-weighted) superior to CT**, enabling differentiation of adrenal masses; **flow-related enhancement** evaluates arteries and veins.
- Usually the **transverse plane at slice thickness ≤3 mm**.
- **T1**: uniform intermediate signal, **slightly lower than liver and renal cortex** (Fig. 145.12C). **T2**: gland **hard to distinguish from retroperitoneal fat**, because intracellular lipid gives similar signal.

Ultrasonography

- Performed mainly to identify abnormalities; normal glands are **hypoechoic** (Fig. 145.12D).
- Limited by retroperitoneal fat, body habitus, the glands' small size and **respiratory motion**.
- Principal use: **differentiating solid from cystic adrenal masses**.

Angiography

- **Adrenal venous sampling**: blood for metabolic assay — **functional information**.
- **Inferior vena cavography**: evaluates the **intraluminal component of an adrenal malignancy** when CT, MRI and US are equivocal.
- **Adrenal arteriography is rarely required**: CT and MRI nearly always supply the morphology needed for surgical planning.

IMAGING MODALITIES COMPARED

Modality	Strength	Limitation	Distinctive role
CT	Best spatial resolution; fat outlines gland	Morphology only	Most widely used
MRI	Best contrast resolution; multiplanar; no radiation	Spatial resolution slightly below CT; gland ≈ fat on T2	Mass characterisation; radiation concern
US	Identifies abnormalities; normal gland hypoechoic	Fat, habitus, small size, respiration	Solid versus cystic

Modality	Strength	Limitation	Distinctive role
Angiography	Venous sampling gives function	Arteriography rarely needed for morphology	Venous sampling; cavography for intraluminal tumour

Concept — Lipid is both ally and confound: *surrounding fat makes the gland conspicuous on CT, while its own intracellular lipid makes it resemble fat on T2 MRI; cross-sectional imaging gives morphology, and only venous sampling gives function.*

CONCLUSION

- The adrenals combine unique anatomic, embryological and histological features; their **transformation from fetal life to adulthood is unlike any other organ.**
- Understanding their **vascular and anatomic relationships is critical** for surgery on the adrenals and nearby organs; advances in imaging improve characterisation of normal and pathological anatomy.

KEY POINTS: ADRENAL GLANDS

- Embryologically the adrenals are **distinct from the kidney**: in renal agenesis the adrenal remains orthotopic, although its shape may be altered.
- The gland comprises **cortex and medulla**; the cortex has three zones — **ZG, ZF and ZR**.
- The medulla receives **preganglionic sympathetic input** that releases catecholamines from chromaffin cells.
- Arterial supply: branches of the **inferior phrenic artery, aorta and renal artery**.
- Venous drainage differs by side: **right into the IVC, left into the left renal vein**.